Zeran Ni (Simon)

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Education

New York University, Center For Data Science, College of Arts and Science *Sep 2023 – Dec 2026*
Major: **Computer Science and Data Science** GPA: **3.925/4.0**
Relevant Courses: Data Structures and Algorithms, Fundamentals of Machine Learning, Software Engineering.

Experience

Research Intern *New York, NY*
RiskEcon® Lab, Courant Institute of Mathematical Sciences, NYU *Dec 2025 – Present*

- Built a Pandas pipeline joining **70K+** hourly load records across 8 ERCOT weather zones with county weather data, reconciling identifiers across four datasets and normalizing timestamps across DST transitions.
- Implemented second-order response surface models of daily load on heating/cooling degree days (HDD/CDD) per zone with NumPy least squares, fitting in natural rather than coded units to avoid collinearity and reaching a preliminary adjusted R^2 above **0.92** in 7 of 8 zones.

Class Tutor and Grader *New York, NY*
Department of Computer Science, New York University *Jan 2024 – Dec 2025*

- Led **10** hours of tutoring and office hours weekly for **200+** students in Python, OOP, and debugging.
- Graded **11** programming assignments for ∼**180** students, delivering detailed and constructive feedback.

Projects

LLM-Assisted Autograder for Introductory CS Course at NYU *GitHub Repository*

- Built an LLM-assisted grading tool used on 11 assignments for 100+ students, cutting grading time by over **50%** by pairing automated tests with GPT evaluation (LangChain) of code quality and style.
- Engineered an evaluation pipeline that runs submissions, validates output with regex and table parsers, and flags edge cases for human review, keeping a grader in the loop for uncertain results.
- Scoped the tool around the instructor's needs: modular per-assignment graders that test whole scripts, so work could be graded before students learned functions, which Gradescope's function-call tests could not support.
- Technologies Used: `Python, OpenAI API, LangChain, Bash, Regular Expression, Git.`

Screenshot2Meet *Chrome Web Store* | *GitHub Repository*

- Built a TypeScript Chrome extension and Python FastAPI REST service that read calendar screenshots with OpenCV and PaddleOCR and fill availability directly into When2meet, a third-party site with no public API, handling multilingual weekday labels and timezone-aware slot planning.
- Troubleshot production reliability after deploying the service in Docker on Fly.io, tracing crashes to out-of-memory kills and slow requests to CPU throttling, and fixed both by tuning memory, CPU, and OCR threads.
- Developed with Claude Code under an approve-before-implement workflow, reviewing each AI-proposed change, and backed the API with 114 Pytest tests, per-IP rate limiting, and request validation.
- Technologies Used: `Python, FastAPI, TypeScript, OpenCV, PaddleOCR, Pytest, Docker, Fly.io.`

Book Exchange *Live Demo* | *GitHub*

- As primary developer on a 3-person team, designed the Flask backend and built the HTML/CSS and JavaScript front end of a full-stack marketplace for students to sell, loan, or donate textbooks, deployed on Vercel.
- Implemented authentication and authorization with Flask-Login sessions and Werkzeug password hashing, restricting edits and deletions of posts and their Cloudinary-hosted images to each post's owner.
- Designed lazy deletion, a single $O(1)$ delete instead of an $O(n)$ sweep across all users' liked posts, with stale references pruned on each user's next login.
- Technologies Used: `Python, Flask, Flask-Login, MongoDB, HTML/CSS, JavaScript, Vercel.`

Technologies

Languages: Python (Primary), Java (Proficient), JavaScript, TypeScript, SQL, C, HTML5/CSS3
AI & LLMs: OpenAI API, LangChain, Claude Code (AI-assisted development), PyTorch, Scikit-learn
Backend & Data: REST APIs, FastAPI, Flask, PostgreSQL, MongoDB, Pandas, NumPy
Cloud & DevOps: Docker, GitHub Actions (CI/CD), Pytest, Git, Linux, Fly.io, DigitalOcean, Vercel